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BLDC and PMSM

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BSc in power & elecrtonics

Alexandria, March 2025

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FOC 2025

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Acknowledgements

Writing Guidance

In the *Acknowledgment* section, express your gratitude to those who helped and supported your work. Start by thanking your advisors, mentors, or supervisors who provided guidance and expertise. Mention any colleagues, classmates, or team members who contributed to discussions or offered assistance. You can also acknowledge specific organisations, institutions, or funding sources that supported your research or work. Lastly, include any personal acknowledgments for family or friends who offered encouragement and moral support during the project. Keep this section sincere, concise, and professional.

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Abstract

Writing Guidance

In the *Abstract* section, provide a concise summary of your project, highlighting the key points. Begin with a brief statement of the problem or objective, followed by a description of your approach or methodology. Summarise the main results or findings, emphasising their significance or implications. Conclude with a sentence or two on the overall contribution or impact of your work. Keep the abstract clear and focused, ideally within 150-250 words, to give readers a quick understanding of your research and its importance.

Keywords: Keyword A, Keyword B, Keyword C.

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Glossary

Latex Typesetting system commonly used for the production of scientific and mathematical documents due to its powerful handling of formulas and references. It is widely used for the preparation of documents such as research papers, theses, reports, and articles. (*p. 35*)

Mathematics Mathematics is what mathematicians do. (*p. 35*)

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Acronyms

GCD Greatest Common Divisor. (*p. 35*)

LCM Least Common Multiple. (*p. 35*)

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1

MOSFETs : SI, SIC

Official Repository: *GitHub Repository*

1.1 MOSFET Switching Characteristics and Parasitic Effects

1.1.1 Abstract

This analysis presents a comprehensive examination of MOSFET switching behavior, with particular focus on parasitic components, transient responses, and performance under extreme operating conditions. The study begins with a fundamental overview of MOSFET technology and progresses through detailed switching operation analysis.

1.1.2 Introduction

Metal Oxide Semiconductor Field Effect Transistors (MOSFETs) represent a cornerstone technology in modern electronics.
They are particularly essential in high-frequency and high-efficiency switching applications. The evolution of this technology spans several decades:

- 1930: Initial FET technology development
- 1950s: Introduction of bipolar transistors
- Mid-1970s: First commercial power MOSFETs
- Present day: Widespread integration in devices from microprocessors to discrete power applications

1.1.3 MOSFET Technology

Fundamental Characteristics

MOSFETs operate as charge-controlled devices, exhibiting several key characteristics:

- Output current proportional to semiconductor charge established by control electrode

- Theoretical switching speeds of 20-200 picoseconds
- No continuous ON current requirement
- Negligible drive current in ON state

Resistive Properties

MOSFETs demonstrate distinct resistive behavior:

- Linear relationship between drain-source voltage drop and semiconductor current
- Characterized by $R_{DS(on)}$ parameter (on-resistance)
- Temperature-dependent resistance characteristics

1.1.4 MOSFET Construction

Physical Structure

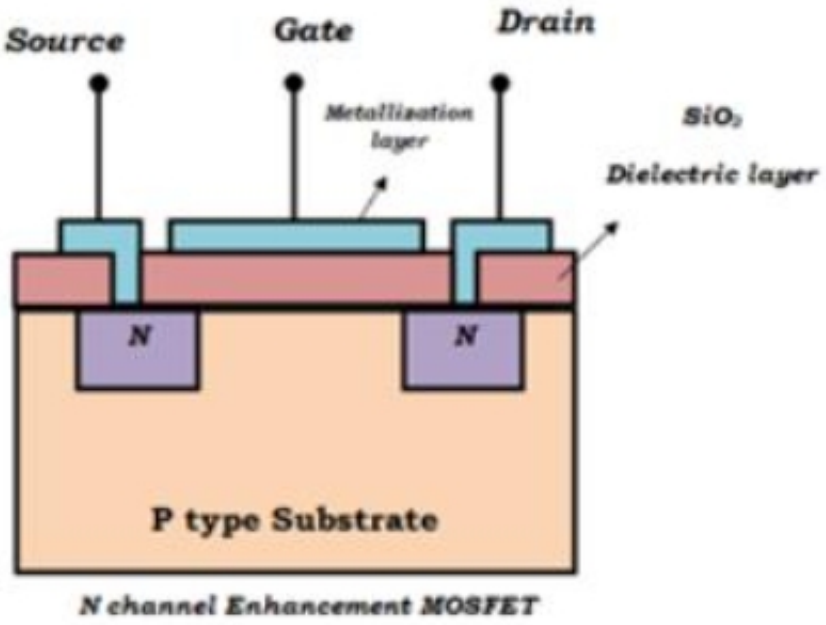


Figure 1.1: *N channel Enhancement MOSFET*

Key components include:

- Metallic gate terminal
- SiO_2 insulation layer (dielectric)
- Three main terminals:
  - Source (S)
  - Gate (G)
  - Drain (D)
- Substrate (body) - internally connected to source

1.1.5 MOSFET Switching Model

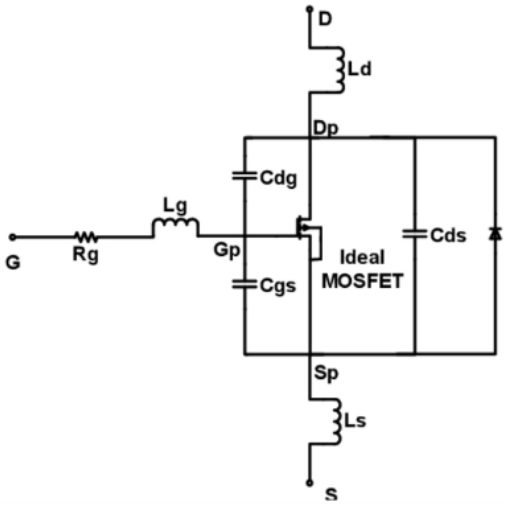


Figure 1.2: *MOSFET Switching Model with Parasitic Components*

Parasitic Components

Critical parasitic elements affecting switching performance:

- Capacitive elements:
  - Gate-source capacitance (C_{gs})
  - Gate-drain capacitance (C_{gd})
  - Drain-source capacitance (C_{ds})
- Inductive elements:
  - Gate inductance (L_g)
  - Source inductance (L_s)
  - Drain inductance (L_d)
- Resistive elements:
  - Internal gate resistance ($R_{g,int}$)
  - Channel resistance (R_{ch})

Model Implications

The switching model demonstrates:

- Interaction between parasitic components
- Impact on switching speed and efficiency
- Critical paths for optimization
- Limitations in high-frequency operation

1.2 MOSFET Critical Parameters

Practical MOSFET switching times (10-60 ns) significantly exceed theoretical switching times (50-200 ps) by two to three orders of magnitude. The switching model reveals three critical capacitors between device terminals, determining switching performance through voltage change rates.
Key capacitive relationships:

$$C_{gd} = C_{rss}$$
$$C_{gs} = C_{iss} - C_{gd}$$
$$C_{ds} = C_{oss} - C_{gd}$$

For voltage-dependent capacitors (C_{gd} and C_{ds}), average capacitances are calculated:

$$C_{gd,avg} = 2 \cdot C_{rss,spec} \cdot \sqrt{\frac{V_{ds,spec}}{V_{ds,off}}}$$
$$C_{oss,avg} = 2 \cdot C_{oss,spec} \cdot \sqrt{\frac{V_{ds,spec}}{V_{ds,off}}}$$

1.2.1 Parasitic Effects

- **Gate Inductance** (L_g):
  - Generates voltage spikes during rapid current changes
  - Forms LC tank with gate-source capacitance
  - Can cause unwanted oscillations
- **Source Inductance** (L_s):
  - Affects both channel current and gate drive
  - Mitigated through Kelvin source connection
- **Drain Inductance** (L_d):
  - Causes ringing with source inductance
  - Controlled through gate resistance and snubber circuits

1.3 MOSFET Switching

1.3.1 Turn-on Procedure

Four distinct intervals characterize turn-on:

1. Input capacitance charging (0V to V_{TH})
2. Gate voltage rise (V_{TH} to $V_{gs,Miller}$)
3. Miller plateau region
4. Final gate voltage rise ($V_{gs,Miller}$ to V_{DRV})

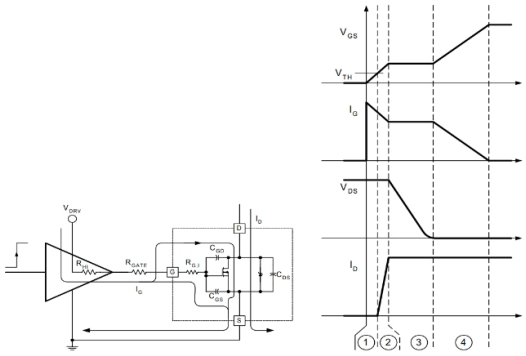
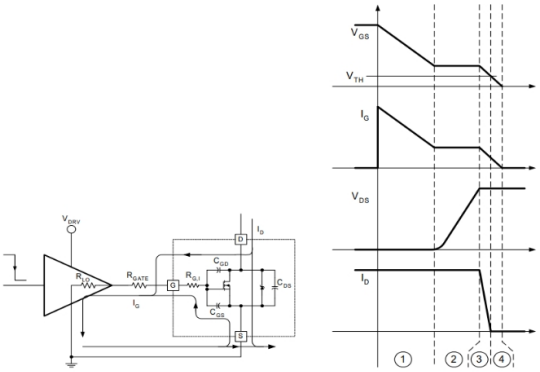


Figure 1.3: *MOSFET Turn-on Waveforms*

1.3.2 Turn-off Procedure

The description of the turn-off procedure for the MOSFET transistor is backtracking the turn-on step



$$V_{gs,Miller} = V_{TH} + \frac{I_d}{g_{fs}}, \text{ where } g_{fs} = \frac{dI_d}{dV_{gs}}$$

1.4 Power Loss Analysis

1.4.1 Gate Drive Losses

Gate drive power loss:

$$P_{gate} = V_{DRV} \cdot Q_g \cdot f_{sw}$$

Driver power loss:

$$P_{driver} = P_{gate} \cdot \frac{R_{Gdriver,out}}{R_{Gdriver,out} + R_G}$$

With some gate drivers, it's easy to get the output resistance of it but, with others, the output resistance is not constant. So, it is safe to assume that with low-value gate resistors ($< 5\,\Omega$) most gate drive losses are dissipated in the driver. If RG is sufficiently large to limit Ig below the output current capability of the bipolar driver, the majority of the gate drive power loss is then dissipated in RG.

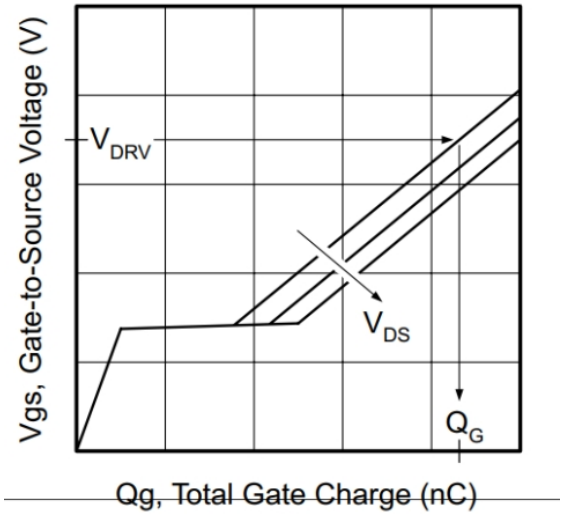
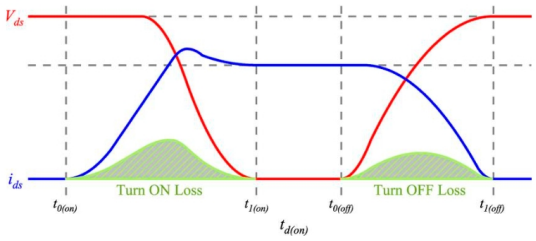


Figure 1.4: *Total Gate Charge*

1.4.2 Switching Losses

In addition to the gate drive power loss, the transistors accrue switching losses due to high current and high voltage being present in the device simultaneously for a short period. In order to ensure the least amount of switching losses, the duration of this time period must be minimized. This condition is limited to intervals 2 and 3 of the switching transitions in both turn-on and turn-off operations. These time intervals correspond to the linear operation of the device when the gate voltage is between VTH and Vgs,Miller, causing changes in the current of the device and to the Miller plateau region when the drain voltage goes through its switching transition.



Gate drive currents:

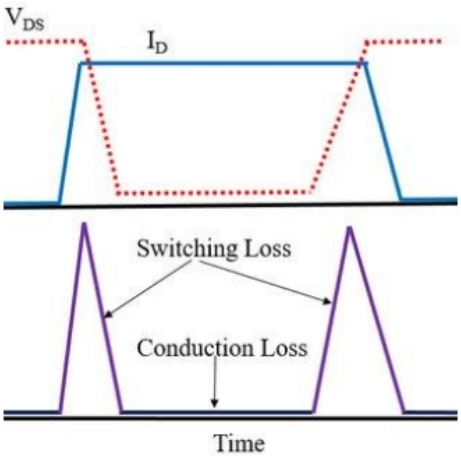
$$I_{G2} = \frac{V_{DRV} - 0.5(V_{gs,Miller} + V_{TH})}{R_{Gdriver,out} + R_g + R_{G,int}}$$

$$I_{G3} = \frac{V_{DRV} - V_{gs,Miller}}{R_{Gdriver,out} + R_g + R_{G,int}}$$

Total switching loss:

$$P_{sw} = \frac{1}{2} \cdot V_{DS,off} \cdot I_L \cdot (t_{on} + t_{off}) \cdot f_{sw}$$

1.4.3 Conduction Losses



$$P_{cond} = I_d^2 \cdot R_{ds,on}$$

1.4.4 Optimal Gate Resistance

An optimum value of the gate resistor can be obtained from the datasheet, the gate resistor used in the tests can be considered as a good starting point to try and start iterations till find the most suitable gate resistor that can get the desired switching speed and acceptable EMI noise. Smaller resistor values will result in an overshoot in the gate drive voltage waveform but also result in a faster turn-on speed. Higher resistor values will underdamp the oscillation and extend the switching times.

$$R_{gate,opt} = 2 \cdot \sqrt{\frac{L_s}{C_{iss}}} - (R_{Gdriver,out} + R_{g,int})$$

1.5 Ground-Referenced Gate Drive

1.5.1 PWM Direct Drive

Primary limitations include:

- Restricted drive current capability
- Maximum current typically below 1A peak

1.5.2 Bipolar Totem-Pole

Advantages:

- Cost-effective solution
- Better handling of current spikes
- Improved PWM controller operating conditions

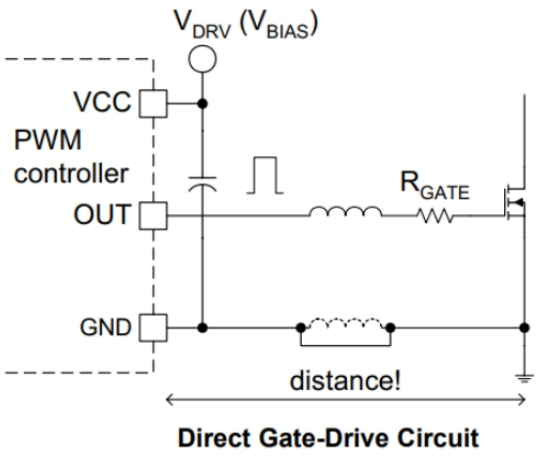


Figure 1.5: *Direct Gate-Drive Circuit*

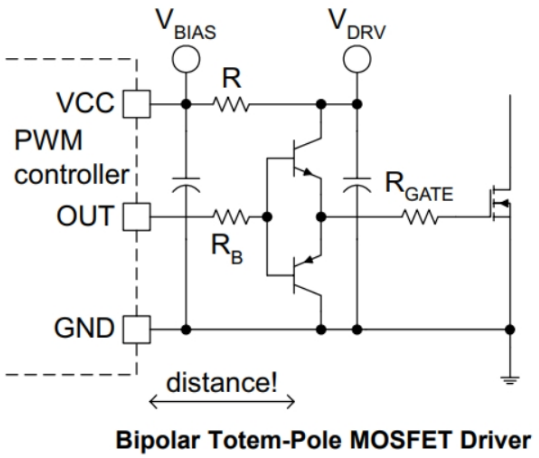


Figure 1.6: *Bipolar Totem-Pole MOSFET Driver*

1.6 Speed-Enhancement Circuits

1.6.1 Turn-Off Diode

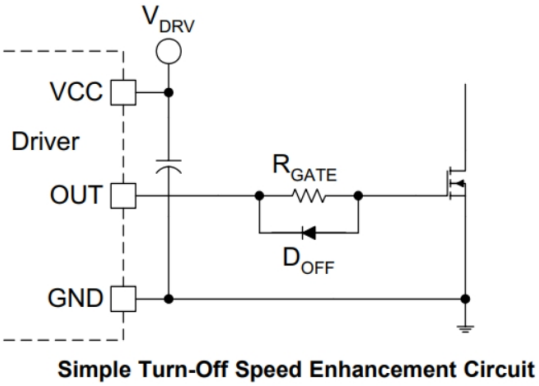


Figure 1.7: *Simple Turn-off Speed enhancement Circuit*

1.6.2 PNP Turn-off Circuit

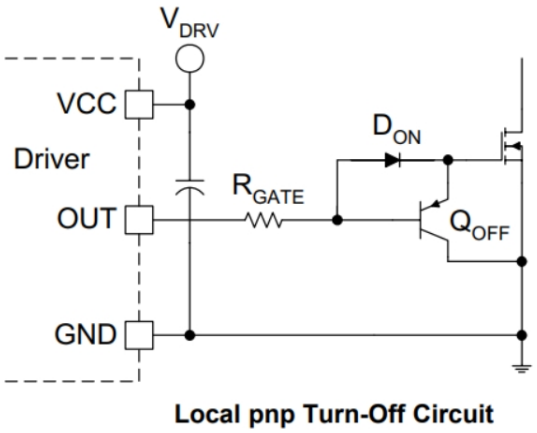


Figure 1.8: *Local PNP Turn-off Enhancement Configuration*

Key features:

- Local gate-source shorting during turn-off
- Reduced driver power dissipation
- Non-saturating operation
- Limitation: Cannot achieve complete 0V gate voltage

1.6.3 NPN Turn-off Circuit

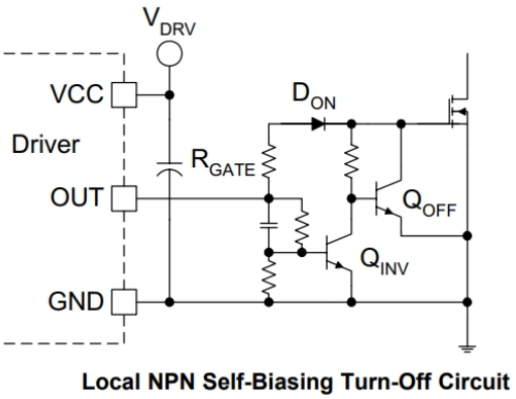


Figure 1.9: *Local NPN Self-Biasing Turn-off Circuit*

Advantage: Better gate voltage hold-down compared to PNP version

1.6.4 NMOS Turn-off Circuit

Features:

- Rapid switching capability
- Complete gate discharge to 0V
- R_{GATE} serves dual purpose:

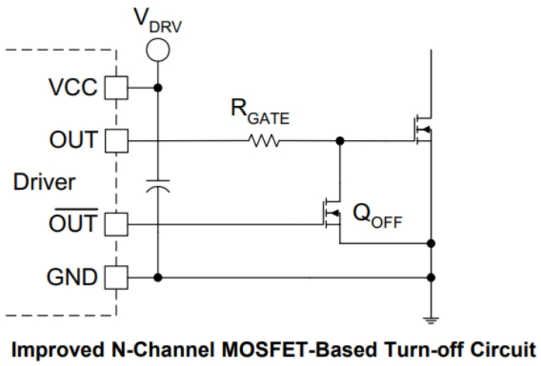


Figure 1.10: *Improved NMOS-Based Turn-off Circuit*

  - Controls turn-on speed
  - Prevents shoot-through currents
- Additional C_{OSS} consideration

1.7 High-side Gate Drives

1.7.1 Bootstrap Gate-Drive Technique

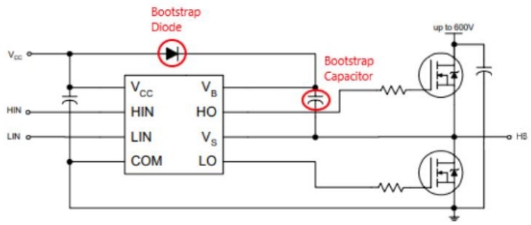


Figure 1.11: *Bootstrap Gate Drive Configuration*

Operation sequence:

1. Low-side FET ON phase:

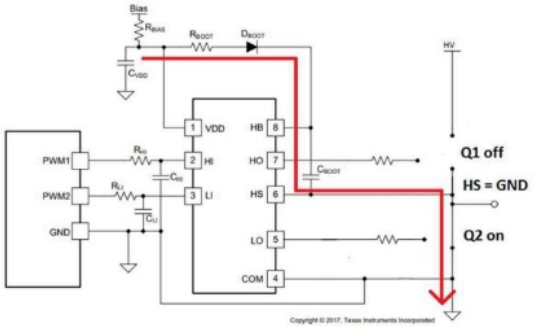


Figure 1.12: *Bootstrap Charging Path*

- Bootstrap capacitor charging through diode

- High-side FET OFF

2. High-side FET ON phase:

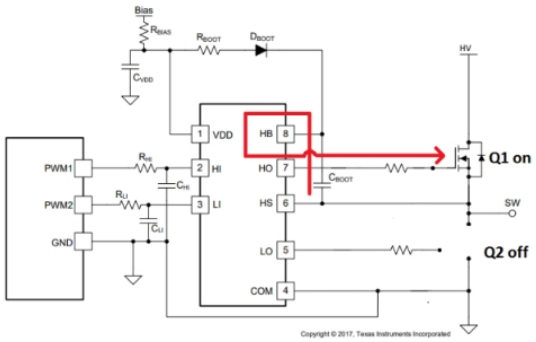


Figure 1.13: *Bootstrap Capacitor Discharging Path*

- Bootstrap capacitor discharge
- Voltage delivery to high-side FET

1.8 Transformer-Coupled Gate Drives

1.8.1 Single-Ended Configuration

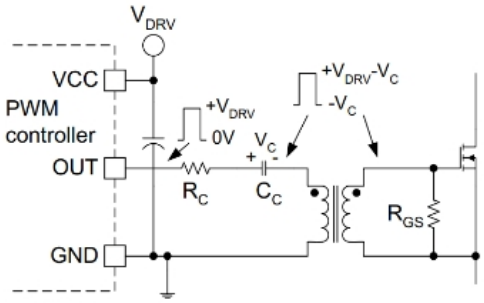


Figure 1.14: *Single-Ended Transformer-Coupled Gate-Drive*

Coupling capacitor voltage:

$$V_C = D \cdot V_{DRV}$$

Damping resistor criterion:

$$R_C \geq 2\sqrt{\frac{L_M}{C_C}}$$

Here, a second coupling capacitor (CC2) and a simple diode clamp (DC2) are used to restore the original gate drive amplitude on the secondary side of the transformer. If a larger negative bias is desired during the off time of the main switch, a Zener diode can be added in series with the diode.

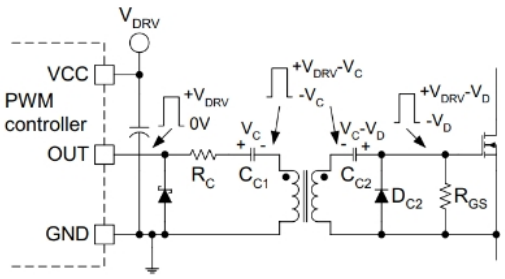


Figure 1.15: *DC Restore Circuit in Transformer-Coupled Gate Drive*

1.8.2 Double-Ended Configuration

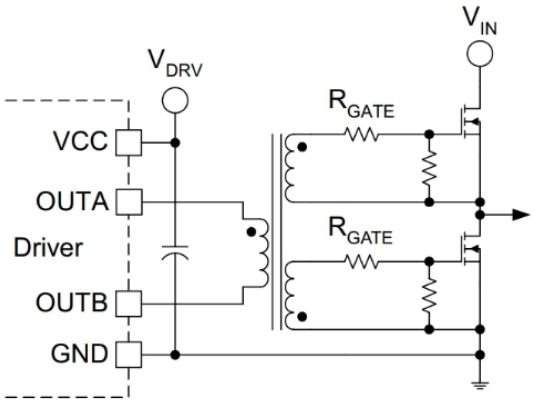


Figure 1.16: *Push-Pull Type Half-Bridge Gate Drive*

Advantages:

- Suitable for high-power applications
- Symmetrical gate drive voltage
- No AC coupling requirement
- Zero average voltage across primary winding

1.9 Practical Design Considerations

1.9.1 Layout Guidelines

- Minimize gate loop inductance
- Separate power and signal grounds
- Optimal component placement
- Proper thermal management

1.9.2 Protection Features

- Over-voltage protection
- Under-voltage lockout

- Short-circuit protection
- Thermal shutdown

1.10 Conclusion

The comprehensive analysis of MOSFET switching behavior reveals the critical importance of:

- Parasitic component management
- Proper gate drive design
- Speed-enhancement techniques
- Protection mechanisms

These factors significantly impact switching performance and overall system reliability.

2
3ph Inverter

Official Repository: GitHub Repository

Welcome to the *IPLeiria Thesis* template! Thank you for choosing it for your dissertation, reports, or other projects. This template reflects countless hours of development and learning, and I hope you enjoy using it as much as I enjoyed creating it. This chapter introduces the motivation behind the template and guides you through the initial steps to start using it. In Chapter 9, you will find a comprehensive guide to fully utilise the template, and Chapter 10 provides a concise LaTeX tutorial to help you maximise its capabilities. Happy reading and future writing!

2.1 Motivation

I've known LaTeX since 2020, and since then, I have been using it constantly for a wide variety of purposes. Over time, I've used and reviewed more than a hundred templates, and there is always something missing. *Always*. If a template is powerful – *i.e.*, it has many options and is highly customisable – it's often not well-organised. Conversely, if it is well-organised, it usually lacks flexibility and customisation options. Some templates even compile with numerous errors and warnings. Most importantly, the majority are not very user-friendly. For these reasons, I decided to create my own template for theses and reports, specifically tailored for the Polytechnic University of Leiria. Throughout this project, I focused on achieving four key goals: i) creating a template that is organised and well-structured in terms of file organisation, ii) ensuring it is clean yet aesthetically pleasing and professional, iii) making it customisable to suit different needs, and iv) designing it to be user-friendly, especially for newcomers.

2.2 Getting Started

To start using this template, you first need to know how to use LaTeX. For this, please refer to Chapter 10. Once you are familiar with LaTeX, you will need to either install it

locally or use an online LaTeX editor.
If you prefer an online editor, I highly recommend Overleaf. While Overleaf offers a paid subscription for extended compilation time, this template is specifically designed to compile within the limits of the free subscription plan. To use the template in Overleaf, just refer to official template page and click *Use as Template*. But if you prefer to use a different version of it (which I do not recommend), you can do the following:

1. *Download the desired version from the GitHub repository as a Zip file.*
2. *Login to your Overleaf account.*
3. *In your Project area, click in the menu and then: New Project → Upload Project.*
4. *Upload the Zip file.*
5. *Let Overleaf compile the document.*

If you choose to use a local editor, you must first install LaTeX on your machine. For this, there are several options, but I personally recommend either TeX-Live or MikTeX. After installing LaTeX, you will need to select an editor for writing and editing your documents. To help with this decision, I suggest checking out this helpful post, which provides a comprehensive overview of various editors you can use. Once LaTeX and your editor are set up, simply clone or download the latest version of the template from GitHub and start using it!

> ***Tip***
>
> *Within the official GitHub repository, you will find a Makefile and a Latexmk configuration file, both of which can be used to compile this project. The Makefile utilises* `rubber` *as the compiler and has a dependency of* `inotify-tools`*. Feel free to use whichever option best suits your needs.*

2.3 Getting Help

As a newcomer, you may encounter situations where you want to do something in LaTeX or with this template but aren't sure how. When questions arise, you have several options. First, you can read the wiki available in the GitHub repository for this template. Another great option is the TeX Stack Exchange, an active community that can help with nearly any issue. Of course, Google is always a reliable resource. If all else fails, feel free to contact me directly with any questions about the template. You can reach me at *jose.apareia@gmail.com*.

2.3.1 Issues, Feature Request and Suggestions

If, by any chance, you encounter a bug, have a suggestion, or would like to request a feature, you can submit them via the *Issues* tab in the GitHub repository. Please be as

descriptive as possible when reporting issues, and make sure to provide the appropriate labels to help me triage them effectively.
For feature requests or suggestions, you can either follow the steps mentioned above or, if you prefer, you can implement the feature yourself and submit a pull request. Both pull requests and pushes trigger a GitHub Action that will automatically compile the document. If the compilation fails, the pull request will be automatically rejected. Please keep this in mind and take care when submitting changes.

2.3.2 In-Built Comments, Guidance Texts and Warnings

Within this document template, you may encounter informational text displaying the message "Writing Guidance." These sections are provided solely as guidance to help users understand what content should be included in specific sections. They are not related to the LaTeX code itself within this template.
While navigating through the template, especially the configuration files, you will notice that everything is thoroughly commented. LaTeX can sometimes be difficult to understand without proper documentation for the packages we are working with. With that in mind, I have made an effort to comment on all the changes I've made. Occasionally, I may forget or deem it unnecessary to comment on simpler changes, but more advanced modifications are always accompanied by detailed comments.
Finally, regarding warnings: please take them into consideration when compiling the document. As you may have noticed, the template is designed to be free of warnings, so please strive to maintain this clean compilation.

2.4 Important Notices

Although this template is specifically tailored for students from all six schools of the Polytechnic University of Leiria, you are welcome to use it if you are from a different institution. It is highly customisable and can be easily modified to suit the needs of other schools. If your school is interested in adopting this as the official template, please contact me first.
If you decide to use this template, please consider acknowledging it in your work. To do so, simply cite it using `\citep{IPLeiriaThesis}`. You can also show your appreciation for this template by giving a star on the GitHub repository. This helps increase its visibility, and you will receive notifications about the latest updates and releases. Either way, acknowledging this work, which involved significant effort and countless hours, would be greatly appreciated.

3
Flyback Converter

Official Repository: GitHub Repository

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3.1 Motivation

I've known LaTeX since 2020, and since then, I have been using it constantly for a wide variety of purposes. Over time, I've used and reviewed more than a hundred templates, and there is always something missing. *Always*. If a template is powerful – *i.e.*, it has many options and is highly customisable – it's often not well-organised. Conversely, if it is well-organised, it usually lacks flexibility and customisation options. Some templates even compile with numerous errors and warnings. Most importantly, the majority are not very user-friendly. For these reasons, I decided to create my own template for theses and reports, specifically tailored for the Polytechnic University of Leiria. Throughout this project, I focused on achieving four key goals: *i*) creating a template that is organised and well-structured in terms of file organisation, *ii*) ensuring it is clean yet aesthetically pleasing and professional, *iii*) making it customisable to suit different needs, and *iv*) designing it to be user-friendly, especially for newcomers.

3.2 Getting Started

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locally or use an online LaTeX editor.
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4
PMSM

Official Repository: GitHub Repository

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4.1 Motivation

I've known LaTeX since 2020, and since then, I have been using it constantly for a wide variety of purposes. Over time, I've used and reviewed more than a hundred templates, and there is always something missing. *Always*. If a template is powerful – *i.e.*, it has many options and is highly customisable – it's often not well-organised. Conversely, if it is well-organised, it usually lacks flexibility and customisation options. Some templates even compile with numerous errors and warnings. Most importantly, the majority are not very user-friendly. For these reasons, I decided to create my own template for theses and reports, specifically tailored for the Polytechnic University of Leiria. Throughout this project, I focused on achieving four key goals: i) creating a template that is organised and well-structured in terms of file organisation, ii) ensuring it is clean yet aesthetically pleasing and professional, iii) making it customisable to suit different needs, and iv) designing it to be user-friendly, especially for newcomers.

4.2 Getting Started

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5 BLDC

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6
FOC PMSM

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6.1 Motivation

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> *Within the official GitHub repository, you will find a Makefile and a Latexmk configuration file, both of which can be used to compile this project. The Makefile utilises* `rubber` *as the compiler and has a dependency of* `inotify-tools`*. Feel free to use whichever option best suits your needs.*

7 Trapezoidal control

Official Repository: GitHub Repository

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7.1 Motivation

I've known LaTeX since 2020, and since then, I have been using it constantly for a wide variety of purposes. Over time, I've used and reviewed more than a hundred templates, and there is always something missing. *Always*. If a template is powerful – *i.e.*, it has many options and is highly customisable – it's often not well-organised. Conversely, if it is well-organised, it usually lacks flexibility and customisation options. Some templates even compile with numerous errors and warnings. Most importantly, the majority are not very user-friendly. For these reasons, I decided to create my own template for theses and reports, specifically tailored for the Polytechnic University of Leiria. Throughout this project, I focused on achieving four key goals: *i*) creating a template that is organised and well-structured in terms of file organisation, *ii*) ensuring it is clean yet aesthetically pleasing and professional, *iii*) making it customisable to suit different needs, and *iv*) designing it to be user-friendly, especially for newcomers.

7.2 Getting Started

To start using this template, you first need to know how to use LaTeX. For this, please refer to Chapter 10. Once you are familiar with LaTeX, you will need to either install it

locally or use an online LaTeX editor.

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>
> *Within the official GitHub repository, you will find a Makefile and a Latexmk configuration file, both of which can be used to compile this project. The Makefile utilises* `rubber` *as the compiler and has a dependency of* `inotify-tools`*. Feel free to use whichever option best suits your needs.*

8
FOC BLDC

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8.1 Motivation

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9

Comprehensive User Guide Instructions for Using the Template

If you plan to use this template, please read this chapter carefully. It provides all the information you need to effectively use the template, including the mandatory modifications (*e.g.*, title, subtitle, author information) and other configurations that, while not highly recommended, are optional. The template comprises various directories and files, including a total of seven distinct directories and dozens of files. Among these, the most important are `IPLeiriaMain.tex` and `IPLeiriaThesis.cls`. Below, Table 9.1 presents the different directories available, along with their descriptions and a check-mark indicating whether you need to access the directory to make changes. Of course, the check-mark indicates that you can make changes to the content, while a hyphen signifies that you should not modify it.

Table 9.1: *Overview of the directory structure in this template.*

Directory	Modifiable	Description
Bibliography	✓	This folder contains the bibliography file used to manage references throughout the document.
Chapters	✓	Individual chapters of the thesis are organised in this directory, making it easy to work on sections separately.
Code	✓	Code examples and relevant scripts are stored here, supporting the content of the thesis.
Configurations	-	All configuration files required for the template, such as layout and style settings, are placed in this directory.
Figures	✓	All figures and images referenced within the document are stored in this folder for easy access and management.
Matter	-	The front matter of the document, including the cover page, copyright statement, and glossary, is assembled in this directory.
Metadata	✓	This folder holds the metadata file, where key document details such as the author, title, and supervisor can be customised.

It is crucial to note that the files are organized according to a specific naming convention, which must be **respected** and **maintained**. The naming convention consists of an ascending two-digit numeric value, followed by a hyphen, and then the file name in capital letters. The name should always aim to be a single word. If more than one word is necessary, they should be separated by a hyphen and capitalised.

> ***Note***
>
> *While Table 9.1 indicates that the Matter directory is not modifiable, two files within that directory should be altered when necessary:* `04-Glossary.tex` *and* `05-Acronyms.tex`*. Although the names are fairly self-explanatory, these files should contain the glossary and acronyms entries, respectively.*

The two files mentioned earlier, `IPLeiriaMain.tex` and `IPLeiriaThesis.cls`, should be used with caution. The main file, as the name suggests, is the master file where you will add the necessary chapters to be included in your work. The class file, on the other hand, requires even more caution, and it is not recommended to alter it.

9.1 Template and Class Options

The first thing you need to do is specify the options within the `IPLeiriaMain.tex` file. How do you do that? It's simple. On the very first line, you will find a `documentclass` command that loads the custom class for this template. In this call, you can pass the options you need. The available options, presented in a key-argument style, are listed in Table 9.2.

Table 9.2: *Options supported by the template.*

Options	Description
school=OPT *estg, esecs, esslei, esad, estm*	**Choosing a school and its corresponding logo.** ⇒ *Default: school=estg* *This option only modifies the school name and the corresponding logo, which will be displayed on the cover and front page.*
language=OPT *portuguese, english*	**Language preference selection.** ⇒ *Default: language=english*
chapterstyle=OPT *classic, modern, fancy*	**Selection of a cover design style.** ⇒ *Default: chapterstyle=classic* *This option modifies the appearance of the chapter, including its title and numbering style. Explore the available styles and apply the one you prefer.*
coverstyle=OPT *classic, bw*	**Choosing a style for the chapter.** ⇒ *Default: coverstyle=classic* *classic* → *Put the cover on in the original red.* *bw* → *Make the cover black and white.*

> ***Warning***
>
> *If any option or argument is passed and it does not exist, an error or a warning will be raised, respectively. Please pay attention to warnings and strive to keep the template and/or your document free of warnings.*

After setting the options within the main class, you're ready to proceed with the metadata customisation. How can you do that? Please refer to Section 9.2.

9.2 Metadata Customisation

While options like language and school can be passed as arguments to the main class, other options, such as author and title, need to be defined manually. Since this template supports a wide range of metadata options, a dedicated file is provided for this purpose. The file is located at `Metadata/Metadata.tex` and contains the metadata variables you can modify, along with comments explaining each variable and whether it is mandatory for the document. To omit a variable, simply comment it out. Below, Table 9.3 presents all the available metadata variables, their GET command, and whether they are mandatory. The GET command is important if you want to automatically retrieve the name or information stored in a given variable.

Table 9.3: *Metadata variables within the template.*

Variable	Macro Commands	Mandatory
Title	`\GetTitle`	✓
Subtitle	`\GetSubtitle`	✓
University	`\GetUniversity`	✓
School	`\GetSchool`	✓
Department	`\GetDepartment`	✓
Degree	`\GetDegree`	✓
Course	`\GetCourse`	-
Local and date	`\GetDate`	✓
Academic year	`\GetAcademicYear`	✓
Thesis type (*Dissertation, Project or Internship*)	`\GetThesisType`	✓
First author name	`\GetFirstAuthor`	✓
First author identification	`\GetFirstAuthorNumber`	✓
Second author name	`\GetSecondAuthor`	-
Second author identification	`\GetSecondAuthorNumber`	-
Third author name	`\GetThirdAuthor`	-
Third author identification	`\GetThirdAuthorNumber`	-
Supervisor name	`\GetSupervisor`	✓
Supervisor e-mail	`\GetSupervisorMail`	✓
Supervisor title and affiliation	`\GetSupervisorTitle`	✓

Table 9.3 continued from previous page.

Variable	Macro Commands	Mandatory
Co-supervisor name	`\GetCoSupervisor`	-
Co-supervisor e-mail	`\GetCoSupervisorMail`	-
Co-supervisor title and affiliation	`\GetCoSupervisorTitle`	-
Second co-supervisor name	`\GetSecCoSupervisor`	-
Second co-supervisor e-mail	`\GetSecCoSupervisorMail`	-
Second co-supervisor title and affiliation	`\GetSecCoSupervisorTitle`	-

If, by any chance, **you want to add more options**, please contact me by opening an issue in the official GitHub repository or via the email provided in this document.

9.3 Custom Commands

Within this template, some custom commands are also available for your use. For example, if you are writing your thesis and want to add a to-do note, you can easily insert a block with the option `todo`, as follows: `\begin{block}[todo]`. This will insert a to-do block with a style similar to Markdown. Other available options are: `tip`, `warning`, and `note`. Below is a visual example for each one.

To-Do

This is a to-do block.

Tip

This is a tip block.

Warning

This is a warning block.

Note

This is a note block.

Another custom command that can come in handy is `\myuline{TEXT}`, which creates a visually improved underline. The default LaTeX underline is not ideal because it sits directly below the bottom margin of the word, increasing the line height of the paragraph. This new command provides a better and more stylish underline.

9.4 Custom Chapter Insertion

As stated before, to use this template, you need to do three things: set the appropriate options in the document class (refer to Section 9.1), update the document metadata (refer to Section 9.2), and create and import your custom chapters. To create and import a custom chapter, follow these steps: *i* create a TeX file under the Chapters directory that follows the predefined naming convention, and *ii*) include it in the main file using the command `\include{CHAPTER}`. And voilà, your first chapter is ready!

10

Essential LaTeX Tutorial Fundamentals and Key Concepts

In this chapter, we will introduce the LaTeX working environment, highlighting the basic essentials you will need to produce your thesis. First of all, LaTeX (pronounced "LAY-tek" or "LAH-tek") is a tool for creating professional-quality documents. Unlike *What You See Is What You Get* editors like Microsoft Word, LaTeX uses plain text files containing both content and formatting commands. These files are processed by a TeX engine, which interprets the commands to produce a polished, typeset PDF. This approach lets you focus on your content, leaving the precise formatting and layout to LaTeX and the TeX engine, ensuring a professional result every time. While this chapter will introduce you to some important functionalities of LaTeX , please take the time to learn LaTeX from the beginning. You can always refer to the Overleaf Learn LaTeX series for guidance.

10.1 Citations

We present two distinct approaches for citing entries in the bibliography. The first method involves in-text citations, executed using `\citet{ENTRY}`, while the second method employs `\citep{ENTRY}` for citations within a paragraph. Below is an example demonstrating both usages. It's essential to note that you can cite multiple works within the same citation environment. To achieve this, you should use the following format: `\citep{ENTRY1, ENTRY2, ...}`. It is also possible to cite only the title of the work or the author of the same. To do this, please use `\citetitle{ENTRY}` for title citations and `\citeauthor{ENTRY}` for author citations.

> ***Tip***
>
> *Proper citations play a crucial role in academic writing, serving as the foundation for credibility, transparency, and the advancement of knowledge. They are a fundamental aspect of responsible scholarly writing. Please ensure accurate and appropriate citations.*

Example: A novel signature scheme is introduced, along with an implementation of the Diffie-Hellman key distribution scheme that accomplishes a public key cryptosystem (Elgamal, 1985). According to Elgamal (1985), a new signature scheme that accomplishes a public key cryptosystem is introduced (...) This template was created by José Areia, with the title *Polytechnic University of Leiria: LaTeX Thesis Template*.

10.2 References

Much like citations, it is advisable to employ references in your document for citing crucial elements such as chapters, sections, figures, or tables. To reference these elements, begin by creating a label. This label can be generated using `\label{TEXT}`, and it should be positioned within the element you intend to refer to. Once the element is created, you can utilise `\ref{LABEL}` to generate an in-text reference. **We strongly recommend using** `\autoref{LABEL}`. This command automatically creates a custom link with colour corresponding to the type of element being referred to. For instance, a chapter reference will appear like this: Chapter 11, rather than simply Chapter 11.

> ***Tip***
>
> *Properly referencing elements within the document, such as* ***chapters, sections, figures, tables, or listings****, is crucial.*

10.3 Glossary and Acronyms

The document includes both a glossary and an acronym list, accessible at the beginning of the document. You can create a new entry in either the `Matter/02-Glossary` or `Matter/03-Acronyms` sections, depending on the type of entry you intend to add. Once the entry is created, you can reference it using `\gls{ENTRY}` for glossary entries. For acronym entries, there are two ways to reference them. The first method, `\acrfull{ENTRY}`, should be used the first time the acronym appears in the text as it automatically provides the definition in-text. Subsequently, to refer to the acronym without repeating its meaning, use `\acrshort{ENTRY}`.

Example: Utilising Latex for Mathematics is essential (...). It is advisable to seek both the Greatest Common Divisor (GCD) and Least Common Multiple (LCM) because (...). Subsequently, with the aid of GCD and LCM, we can (...).

10.4 Figures

In LaTeX, integrating figures is a straightforward process. To insert them, you should utilise the environment `\begin{figure}`. You can customise the `width` parameter according to your requirements, but it is crucial to select a high-quality figure when inserting it into your documents. It is equally crucial to furnish a well-crafted caption. If necessary, consider including citations or references to indicate the figure's origin. The caption environment is denoted as `\caption{TEXT}`. To generate a smaller caption for the Table of Figures, be sure to utilise the format `\caption[SMALL_TEXT]{BIG_TEXT}`. By following the aforementioned tips, we can create a figure as demonstrated in Figure 10.1.

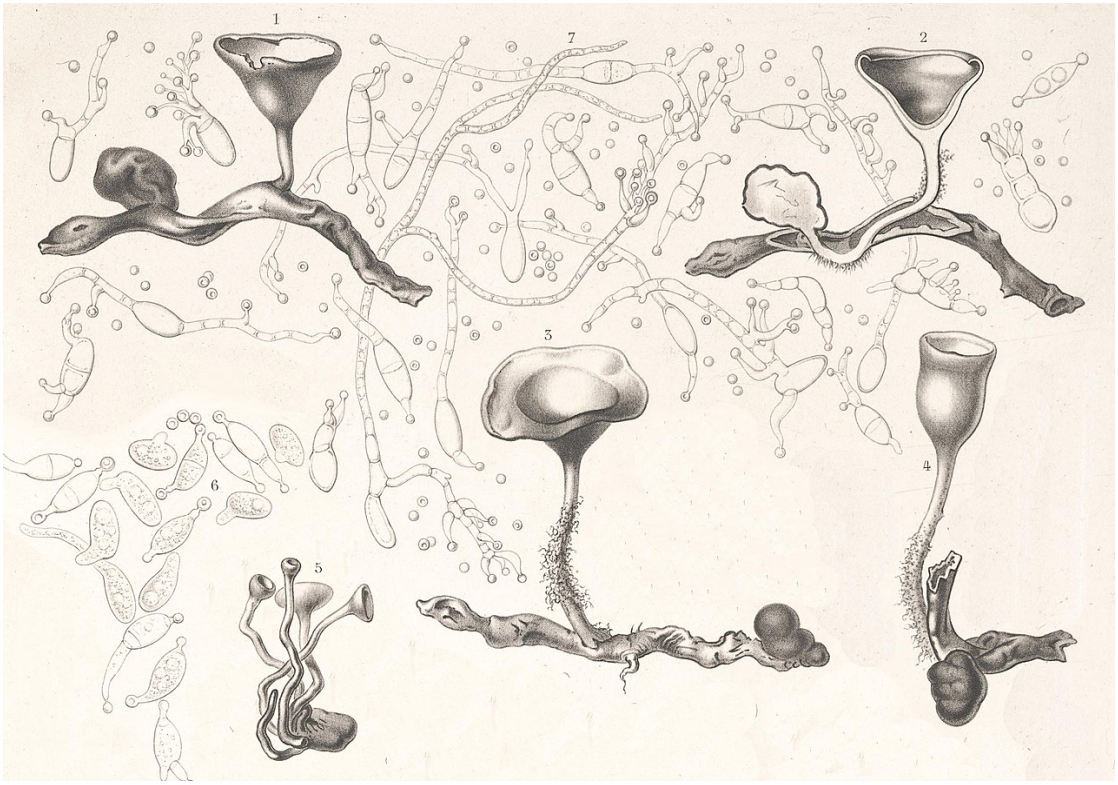
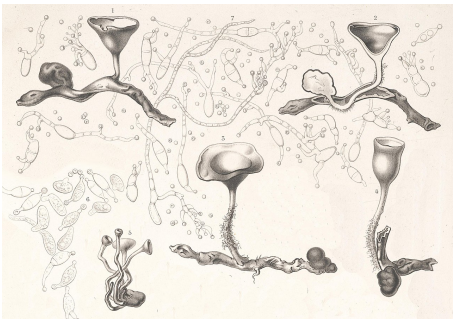


Figure 10.1: *Illustration of the fungus Dumontinia tuberosa by physician, mycologist, and illustrator Charles Tulasne (1816–1884) in the book Selecta Fungorum Carpologia (1861–65). (Name of the original work: Peziza tuberosa parasite on Anemone nemorosa).*

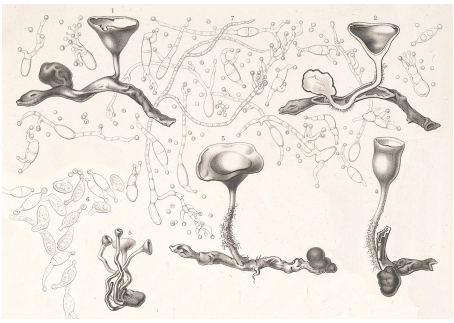
For the purpose of comparing or for other reasons, you can insert side-by-side figures using both the `\begin{figure}` and `\begin{subfigure}` environments. You can also refer to the sub-figure as Figure 10.2a and Figure 10.2b.

10.5 Tables

Tables are vital for presenting findings effectively. This chapter explores techniques for conveying information through tables using various template environments. Defining tables in LaTeX seems complex, but this template simplifies the process.



(**a**) *Caption for figure 1.*



(**b**) *Caption for figure 2.*

Figure 10.2: *Overall caption of the figure.*

> ***Tip***
>
> *Prior to showcasing the different table environments, it's crucial to note that each one must be enclosed within a* `\begin{table}` *environment. Additionally, it is recommended to utilise the* `[!htpb]` *float options for improved document placement.* ***This advice should be taken into consideration when positioning figures as well.***

10.5.1 Tabular Environment

The conventional `\begin{tabular}` environment enables you to create a simple yet elegant table. Table 10.1 is generated using a centering environment for added emphasis. It also incorporates the `booktab` configuration for a more sophisticated table style.

Table 10.1: *A table showcasing the usage of the tabular environment.*

Header 01	**Header 02**	**Header 03**
Lorem Ipsum	Pharetra Dolor	✓
Amet Consectetuer	Curabitur Aliquet	-
Praesent Mauris	Praesent Libero	✓

10.5.2 Tabularx Environment

Employ the `\begin{tabularx}` package to construct a table featuring automatically expanding multi-columns. To achieve this automatic behaviour for multi-columns, you can use the following environment: `\begin{tabularx}{\textwidth}{lX}`, where `X` is the column that will function as a multi-column. Table 10.2 showcases the usage of the `begin{tabularx}` environment.

10.5.3 Longtable Environment

At times, when dealing with exceptionally lengthy tables, it becomes necessary to split them across multiple pages. In LaTeX, this can be achieved using the `\begin{longtable}` environment. This environment is slightly more complex than others, as you need to

Table 10.2: *A table showcasing the usage of the tabularx environment.*

Header 01	**Header 02**
Foo Bar Baz	Quisque cursus, metus vitae pharetra auctor, sem massa mattis sem, at interdum magna augue eget diam.
Ipsum Dolor	Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Curabitur aliquet quam id dui.
Dolor Sit	Phasellus condimentum elementum justo, quis interdum est sagittis ac. Vestibulum non arcu sit amet justo lobortis semper.
Amet Consectetuer	Integer nec odio praesent libero sed cursus ante dapibus diam sed nisi vestibulum non arcu.
Consectetuer Adipiscing	Nulla quis sem at nibh elementum imperdiet. Duis sagittis ipsum. Praesent mauris.

define the header twice: once for the initial appearance of the table and again for when the table spans additional pages. This repeated header ensures the reader can correctly identify the columns on subsequent pages. Feel free to consult Table 10.3 for a detailed demonstration of how the `longtable` environment works.

Table 10.3: *A table showcasing the usage of the longtable environment.*

Names	**E-Mails**	**Job/Role**
Alice Johnson	alice.johnson@email.com	Project Manager
Bob Thompson	bob.thompson@email.com	Data Analyst
Charlie Davis	charlie.davis@email.com	Marketing Specialist
David Miller	david.miller@email.com	QA Tester
Emily White	emily.white@email.com	Graphic Designer
Frank Martin	frank.martin@email.com	HR Coordinator
Grace Turner	grace.turner@email.com	Financial Analyst
Henry Lee	henry.lee@email.com	System Administrator
Ivy Carter	ivy.carter@email.com	Customer Support
Jack Wilson	jack.wilson@email.com	Frontend Developer
Jane Reed	jane.reed@email.com	UX Designer
Kevin Evans	kevin.evans@email.com	Product Manager
Linda Adams	linda.adams@email.com	Accountant
Mike Hill	mike.hill@email.com	Network Engineer
Nina Garcia	nina.garcia@email.com	Business Analyst
Oliver Smith	oliver.smith@email.com	Sales Representative
Pamela Turner	pamela.turner@email.com	Legal Counsel
Quincy Brown	quincy.brown@email.com	IT Consultant
Rachel Moore	rachel.moore@email.com	Content Writer
Samuel White	samuel.white@email.com	Research Scientist
Amy Harris	amy.harris@email.com	Digital Strategist
Brian Cook	brian.cook@email.com	Operations Manager
Catherine Ross	catherine.ross@email.com	Brand Manager
Daniel Green	daniel.green@email.com	Database Administrator

Table 10.3 continued from previous page.

Names	E-Mails	Job/Role
Emma Taylor	emma.taylor@email.com	Social Media Manager
Felix Carter	felix.carter@email.com	Compliance Officer
Gloria Scott	gloria.scott@email.com	Procurement Specialist
Harold Bennett	harold.bennett@email.com	DevOps Engineer
Isla Cooper	isla.cooper@email.com	User Researcher
James Black	james.black@email.com	Mobile App Developer
Katie Brown	katie.brown@email.com	UI Designer
Leo Perez	leo.perez@email.com	Scrum Master
Megan Clark	megan.clark@email.com	Event Coordinator
Nathan Ward	nathan.ward@email.com	Security Analyst
Olivia Harris	olivia.harris@email.com	Corporate Trainer
Paul King	paul.king@email.com	Territory Manager
Queen Foster	queen.foster@email.com	Paralegal
Rebecca Adams	rebecca.adams@email.com	Copy Editor
Steven Martin	steven.martin@email.com	Robotics Engineer

10.5.4 Complex Tables

Creating intricate tables in LaTeX can be a somewhat challenging task. Therefore, we highly recommend using the Table Generator. With this tool, you can design your table with the desired style and then easily copy and paste it into your document. This approach simplifies the process and helps ensure the accurate representation of complex tables in your LaTeX document. However, it's crucial to keep in mind that a table should be easily comprehensible for the reader and should not be overly complex. **The complexity of a table may impede understanding.** For example, Table 10.4 presents a table with intricate details.

10.6 Lists

Creating lists in LaTeX is straightforward, offering various options to suit your needs. You can generate a bullet list using `\begin{itemize}`, or opt for a numbered list with `\begin{enumerate}`. Below is an example with the `\begin{itemize}` environment.

- List entries start with the `\item` command.
- Individual entries are indicated with a black dot, a so-called bullet.
- The text in the entries may be of any length.

As mentioned earlier, you can generate a numbered list using the `\begin{enumerate}` environment. Here is an example:

1. Items are numbered automatically.

Table 10.4: *A table showcasing the usage of the complex tables.*

Component	**Specifications**	
	Characteristic	**Supported**
CPU	Core Count (e.g., 8 Cores)	✓
	Clock Speed (e.g., 3.6 GHz)	✓
	Hyper-Threading	✓
	Integrated Graphics	-
GPU	CUDA Cores (e.g., 5120)	✓
	Base Clock (e.g., 1.5 GHz)	✓
	Ray Tracing Support	✓
	Multi-GPU Support (SLI/CrossFire)	-
Memory	Type (e.g., DDR5, GDDR6)	✓
	Capacity (e.g., 16 GB)	✓
	Memory Bandwidth (e.g., 448 GB/s)	✓
	ECC Support	-
Motherboard Features	PCIe 5.0 Support	✓
	Wi-Fi 6E	✓
	Thunderbolt 4	-

2. The numbers start at 1 with each use of the `enumerate` environment.
3. Another entry in the list.

You can also nest list entries by creating a list inside another list of the same type. Here is an example:

1. First level item
2. First level item
   (a) Second level item
   (b) Second level item
      i. Third level item
      ii. Third level item
         A. Fourth level item
         B. Fourth level item

> ***Tip***
>
> *Please note that the labels change automatically regardless of the environment being the same for every list.* ***This demonstrates that there's no need to worry about changing the environment for something different.*** *However, if desired, you have the flexibility to do so.*

You can also modify the label of your list to something entirely different that suits your needs. To accomplish this, insert a new `\item` and enclose your desired label in square

brackets. For example, `\item[!]` will result in an exclamation point as your new label. Below are some examples of modified labels.

- This is my first point
- Another point I want to make

! A point to exclaim something!

■ Make the point fair and square.
A blank label?

Finally, you can create a description list. Unlike having a bullet point or a numbered label, a description list enables you to use custom descriptions that suit your list. In the example below, there are three `\item` entries: one without a label, and two with descriptions.

Item 1: This is the first item with a description.
Item 2: Another item with a different description.
An item without a specific label.

10.7 Code Listings

At times, you may want to include source code from your programs and applications within your document. To achieve this, you can use two nested environments: `\begin{listing}` to create a listing with both caption and label, and `\begin{minted}` for code highlighting. Listing 10.1 provides an example of a source code in C.

```c
#include <stdio.h>
int main() {
   printf("Hello, World!"); /* printf() outputs the quoted string */
   return 0;
}
```

Listing 10.1: *Hello world in C.*

The code mentioned above was inserted into the document. However, an alternative approach is to input your code from an external file. To do so, you just need to use the command `\inputminted{CODE_LANGUAGE}{FILE}`. Of course, you should place that command inside of the `\begin{listing}` environment. Listing 10.2 illustrates an example of Haskell source code that has been input from an external file.
In some cases, when you simply want to highlight a specific command, it's recommended not to use `listing` or `minted`. Instead, you should utilise the `\verb` command for inline highlighting or the `\begin{verbatim}` environment for longer sections of highlighted code. An example of a lengthy `verbatim` section is provided below, demonstrating how to create a `listing` with an input code:

```
\begin{listing}[!htpb]
```

```haskell
-- Factorial function
factorial :: Integer -> Integer
factorial 0 = 1  -- Base case: 0! = 1
factorial n = n * factorial (n - 1)  -- Recursive case

-- Main function to test the factorial
main :: IO ()
main = do
    putStrLn "Enter a number:"
    input <- getLine
    let number = read input :: Integer
    print (factorial number)
```

Listing 10.2: *Factorial in Haskell.*

```
    \inputminted{CODE_LANGUAGE}{FILE}
    \caption{TEXT}
    \label{TEXT}
\end{listing}
```

Sometimes it is necessary to display longer code that occupies more than one page. For this purpose, please use the environment `\begin{longlisting}`. This environment will easily break your code into multiple pages for better readability without you worrying about the size of your code. An example is shown below in Listing 10.3.

```lisp
(defun factorial (n)
  "Calculate the factorial of a number."
  (if (zerop n)
      (* n (factorial (1- n)))))

(defun fibonacci (n)
  "Calculate the nth Fibonacci number."
  (cond ((zerop n) 0)
        ((= n 1) 1)
        (t (+ (fibonacci (1- n)) (fibonacci (- n 2))))))

(defun gcd (a b)
  "Calculate the greatest common divisor of a and b."
  (if (zerop b)
      a
      (gcd b (mod a b))))

(defun primes-up-to (limit)
  "Return a list of all prime numbers up to LIMIT."
  (let ((primes '()))
    (loop for i from 2 to limit
          unless (some (lambda (p) (zerop (mod i p))) primes)
          do (push i primes))
    (nreverse primes)))
```

```lisp

(defun example-function (x)
  "An example function to demonstrate Lisp capabilities."
  (let ((result (list (factorial x)
                      (fibonacci x)
                      (gcd x 10)
                      (primes-up-to x))))
    (format t "Factorial of ~A: ~A~%" x (factorial x))
    (format t "Fibonacci of ~A: ~A~%" x (fibonacci x))
    (format t "GCD of ~A and 10: ~A~%" x (gcd x 10))
    (format t "Primes up to ~A: ~A~%" x (primes-up-to x))
    result))

(example-function 10)
```

Listing 10.3: *A sample of functions in Lisp.*

10.8 Equations

When writing equations and other mathematical expressions, LaTeX is a powerful and versatile tool. You can enter a formula in inline mode using the environment `\(FORMULA\)` or use `\begin{equation}` to display it in "math mode" with numbering. If you prefer not to display the equation number, you can use the environment `\[FORMULA\]`.

Example: In physics, the mass-energy equivalence is expressed by the equation $E = mc^2$, discovered in 1905 by Albert Einstein. In natural units ($c = 1$), the formula (10.1) expresses the identity:

$$E = m \tag{10.1}$$

Example: Below is a equation – *without numbering* – for the regularised loss function in supervised learning, combining the average prediction loss over the training dataset and an L_2 regularisation term to prevent overfitting:

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^{N} \ell(y_i, f(\mathbf{x}_i; \boldsymbol{\theta})) + \lambda \|\boldsymbol{\theta}\|_2^2$$

Equations can be a bit challenging to create, so we advise using an online editor, like the LaTeX Equation Editor. Simply build your formulas there and copy and paste them into your document, either inline or in a math block, as shown above.

10.9 Footnotes

Sometimes it is important to present information that is not central to the main text in a footnote. In LaTeX this can be easily achieved using the command `\footnote{TEXT}`. The text will appear at the bottom of the page[1].

[1] This is a simple footnote.

If you want to use footnotes within tables, it is best to reconsider, as LaTeX does not provide an easy way to handle them. Instead, you can place a "*" wherever you want the footnote reference to appear. Then, below the table **but before ending the table environment**, place the "*" along with the footnote text. This will create a similar footnote, but it will appear below the table rather than at the bottom of the page.

11

Introduction to the Template: Motivation and First Steps

Official Repository: GitHub Repository

Welcome to the *IPLeiria Thesis* template! Thank you for choosing it for your dissertation, reports, or other projects. This template reflects countless hours of development and learning, and I hope you enjoy using it as much as I enjoyed creating it. This chapter introduces the motivation behind the template and guides you through the initial steps to start using it. In Chapter 9, you will find a comprehensive guide to fully utilise the template, and Chapter 10 provides a concise LaTeX tutorial to help you maximise its capabilities. Happy reading and future writing!

11.1 Motivation

I've known LaTeX since 2020, and since then, I have been using it constantly for a wide variety of purposes. Over time, I've used and reviewed more than a hundred templates, and there is always something missing. *Always*. If a template is powerful – *i.e.*, it has many options and is highly customisable – it's often not well-organised. Conversely, if it is well-organised, it usually lacks flexibility and customisation options. Some templates even compile with numerous errors and warnings. Most importantly, the majority are not very user-friendly. For these reasons, I decided to create my own template for theses and reports, specifically tailored for the Polytechnic University of Leiria. Throughout this project, I focused on achieving four key goals: *i*) creating a template that is organised and well-structured in terms of file organisation, *ii*) ensuring it is clean yet aesthetically pleasing and professional, *iii*) making it customisable to suit different needs, and *iv*) designing it to be user-friendly, especially for newcomers.

11.2 Getting Started

To start using this template, you first need to know how to use LaTeX. For this, please refer to Chapter 10. Once you are familiar with LaTeX, you will need to either install it locally or use an online LaTeX editor.
If you prefer an online editor, I highly recommend Overleaf. While Overleaf offers a paid subscription for extended compilation time, this template is specifically designed to compile within the limits of the free subscription plan. To use the template in Overleaf, just refer to official template page and click *Use as Template*. But if you prefer to use a different version of it (which I do not recommend), you can do the following:

1. *Download the desired version from the GitHub repository as a Zip file.*
2. *Login to your Overleaf account.*
3. *In your Project area, click in the menu and then: New Project → Upload Project.*
4. *Upload the Zip file.*
5. *Let Overleaf compile the document.*

If you choose to use a local editor, you must first install LaTeX on your machine. For this, there are several options, but I personally recommend either TeX-Live or MikTeX. After installing LaTeX, you will need to select an editor for writing and editing your documents. To help with this decision, I suggest checking out this helpful post, which provides a comprehensive overview of various editors you can use. Once LaTeX and your editor are set up, simply clone or download the latest version of the template from GitHub and start using it!

> ***Tip***
>
> *Within the official GitHub repository, you will find a Makefile and a Latexmk configuration file, both of which can be used to compile this project. The Makefile utilises `rubber` as the compiler and has a dependency of `inotify-tools`. Feel free to use whichever option best suits your needs.*

11.3 Getting Help

As a newcomer, you may encounter situations where you want to do something in LaTeX or with this template but aren't sure how. When questions arise, you have several options. First, you can read the wiki available in the GitHub repository for this template. Another great option is the TeX Stack Exchange, an active community that can help with nearly any issue. Of course, Google is always a reliable resource. If all else fails, feel free to contact me directly with any questions about the template. You can reach me at *jose.apareia@gmail.com*.

11.3.1 Issues, Feature Request and Suggestions

If, by any chance, you encounter a bug, have a suggestion, or would like to request a feature, you can submit them via the *Issues* tab in the GitHub repository. Please be as descriptive as possible when reporting issues, and make sure to provide the appropriate labels to help me triage them effectively.
For feature requests or suggestions, you can either follow the steps mentioned above or, if you prefer, you can implement the feature yourself and submit a pull request. Both pull requests and pushes trigger a GitHub Action that will automatically compile the document. If the compilation fails, the pull request will be automatically rejected. Please keep this in mind and take care when submitting changes.

11.3.2 In-Built Comments, Guidance Texts and Warnings

Within this document template, you may encounter informational text displaying the message "Writing Guidance." These sections are provided solely as guidance to help users understand what content should be included in specific sections. They are not related to the LaTeX code itself within this template.
While navigating through the template, especially the configuration files, you will notice that everything is thoroughly commented. LaTeX can sometimes be difficult to understand without proper documentation for the packages we are working with. With that in mind, I have made an effort to comment on all the changes I've made. Occasionally, I may forget or deem it unnecessary to comment on simpler changes, but more advanced modifications are always accompanied by detailed comments.
Finally, regarding warnings: please take them into consideration when compiling the document. As you may have noticed, the template is designed to be free of warnings, so please strive to maintain this clean compilation.

11.4 Important Notices

Although this template is specifically tailored for students from all six schools of the Polytechnic University of Leiria, you are welcome to use it if you are from a different institution. It is highly customisable and can be easily modified to suit the needs of other schools. If your school is interested in adopting this as the official template, please contact me first.
If you decide to use this template, please consider acknowledging it in your work. To do so, simply cite it using `\citep{IPLeiriaThesis}`. You can also show your appreciation for this template by giving a star on the GitHub repository. This helps increase its visibility, and you will receive notifications about the latest updates and releases. Either way, acknowledging this work, which involved significant effort and countless hours, would be greatly appreciated.

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References

Elgamal, T. (July 1985). "A public key cryptosystem and a signature scheme based on discrete logarithms". In: *IEEE Transactions on Information Theory* 31.4, pp. 469–472. ISSN: 1557-9654. DOI: 10.1109/TIT.1985.1057074. URL: https://ieeexplore.ieee.org/document/1057074.

José Areia (Dec. 2023). *Polytechnic University of Leiria: LaTeX Thesis Template*. URL: https://github.com/joseareia/ipleiria-thesis.

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Appendices

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A

Showcasing the First Appendix

Writing Guidance

Appendices contain supplementary material **created by the author** that enhances the reader's understanding of the dissertation while not being essential for following the primary narrative. These sections often include detailed tables, figures, complex calculations, or materials like survey questions and interview transcripts produced in the course of the research. The appendices allow readers to explore the research in greater detail, offering a deeper insight into methods and findings without interrupting the main body of work.

B

Showcasing the Second Appendix

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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Annexes

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L

Showcasing the First Annex

Writing Guidance

Annexes are supplementary sections in a dissertation that provide additional information or external documents not essential to the main arguments but that support or complement the research. Unlike appendices, **annexes generally contain material that was not developed by the author**, such as reports, legal documents, or published datasets from external sources. This information is placed separately to keep the main content concise, allowing readers access to relevant external references without disrupting the dissertation's flow.

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